

Terrestrial Precipitation: 1900-2017 Gridded Monthly Time Series

(Version 5.01)

interpolated and documented by

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STATION DATA SOURCES:

Station data, monthly-total raingage-measured precipitation (P , mm), were compiled from several updated sources including a recent version of the Global Historical Climatology Network dataset ([GHCN2](#)); a version of the Daily Global Historical Climatology Network ([GHCN-Daily](#)) (Menne et al., 2012); an Atmospheric Environment Service/Environment Canada archive; data from the Hydrometeorological Institute in St. Petersburg, Russia (courtesy of Nikolay Shiklomanov); GC-Net data (Steffen et al., 1996); Greenland station records from the Automatic Weather Station Project (courtesy of Charles R. Stearns at the University of Wisconsin-Madison); daily data for India¹ from the National Center for Atmospheric Research (NCAR); Sharon Nicholsons archive of African precipitation data (2001)²; Webber and Willmotts (1998) South American monthly precipitation station records³ (for notes 1, 2, and 3, see the documentation [README file](#)); and daily records from the Global Surface Summary of Day ([GSOD](#)). Station climatologies from Legates and Willmotts (1990) unadjusted (for raingage undercatch) archive also were used as a part of the background climatology (see Spatial Interpolation below). Station P values were not adjusted to reduce raingage undercatch bias.

Some of the P archives contained daily observations and monthly P values were derived from those archives. A monthly station value was calculated from the daily station observations, only when the number of missing daily observations for that month was no more than five. If more than five daily observations were missing, the monthly P value was encoded as missing. In the past, within the GSOD archive, we observed that some of the daily precipitation observations appeared to be unrealistic. To mitigate potentially deleterious influences from unrealistic records, we first applied several filters (similar to filters described by Durre et al., 2010) to the original GSOD daily values. This filtering helped remove many of the unrealistic records, including duplicated months and years.

Many of the available station records were merged to create composite P station-record series. Station records located within 2.5 km of each other were combined and subsequently treated as an individual station record. In order to reduce the influences of outliers, the median of each set of same-month P values was taken as that month's composite station-record value. The location coordinates of each composite station record were taken as the averages of the merged stations' two

geographic coordinates. To check for unusual values in each composite time series, the estimated values were compared with monthly climatological norms estimated by Legates and Willmott (1990). Legates and Willmott (LW) station norms were interpolated to each composite station-record location and the absolute difference between each monthly composite-station value and the estimated monthly LW norm was calculated, over the period of record. For each composite-station record, the interquartile range of monthly differences was estimated. To identify unusual values, the following relationship was evaluated: $dP_i - q50 > F (q75 - q25)$ where dP_i is an absolute difference between an estimated LW climatology value and a composite-station value for month i ; $q50$, $q75$, and $q25$ are the median, 75th and 25th percentiles of the differences for a station, over the period of record. F is a multiplier and it is set to 5.6 for precipitation which reduces the number of outliers (cf, Eischeid et al., 1995). If the difference between dP_i and $q50$ was greater than 5.6 times the interquartile range, the monthly value was considered to be an outlier. Values identified as outliers in this way comprised about 0.35% of the population of P observations and were excluded from the composite-station record. Several P records also contained questionable strings of 0 P values. These were similarly compared with LW climatology values and, if unusual, treated as missing values.

SPATIAL INTERPOLATION:

Station values of monthly total raingage-measured precipitation (P) were interpolated to a 0.5-degree by 0.5-degree latitude/longitude grid, where the grid nodes are centered on the 0.25 degree. Climatologically aided interpolation (CAI) (Willmott and Robeson, 1995) was used to estimate our monthly total precipitation fields. By using a background climatology based on a relatively dense network of stations, CAI can increase the accuracy of spatially interpolated time series of monthly climate variables. For our background climatology, two station climatologies were merged. The first was calculated at those of our precipitation time-series stations which had at least ten years of observations for each month. The second was the monthly station P (raw raingage) climatology of Legates and Willmott (1990). Only those Legates and Willmott stations which were not collocated with our own climatology were included in the background climatology for CAI. A monthly P value at each time-series station was differenced from our climatologically averaged P for that month, which was available at or was interpolated spatially to the time-series station location. Traditional interpolation (Willmott et al., 1985) then was performed on the monthly station differences to obtain a gridded difference field. Finally, each gridded monthly difference field was added to the gridded estimates of the month's climatology at the corresponding set of grid points.

Traditional interpolation was accomplished with the spherical version of Shepard's algorithm, which employs an enhanced distance-weighting method (Shepard, 1968; Willmott et al., 1985). The number of nearby stations that influenced a grid-node estimate was increased to an average of 20, from an average of 7 in earlier applications. This resulted in smaller cross-validation errors (see below) and visually more realistic precipitation fields. A more robust neighbor finding algorithm, based on spherical distance, also was used.

SPATIAL CROSS VALIDATION:

To indicate (roughly) the spatial interpolation errors, station-by-station cross validation was employed (Willmott and Matsuura, 1995). One station was removed at a time, and the precipitation value was then interpolated to the removed station location from the surrounding nearby stations. The difference between the real station value and the interpolated value is a local estimate of interpolation error. After each station cross validation was made, the removed station was put back into the network. To reduce network biases on cross-validation results, absolute values of the errors at the stations were interpolated to the same spatial resolution as the precipitation field.

ARCHIVE STRUCTURE:

precip_2017.tar.gz: Monthly total precipitation for the years 1900-2017 interpolated to a 0.5 by 0.5 degree grid resolution (centered on 0.25 degree). The format of each record is:

Field	Columns	Variable	Fortran Format
1	1 - 8	Longitude (decimal degrees)	F8.3
2	9 - 16	Latitude (decimal degrees)	F8.3
3-14	17 - 112	Monthly Total Precipitation (mm)	12F8.1

precip_cv2017.tar.gz: Cross-validation errors (absolute values) associated with precipitation for the years 1900-2017 interpolated to a 0.5 by 0.5 degree grid resolution. The format of each record is:

Field	Columns	Variable	Fortran Format
1	1 - 8	Longitude (decimal degrees)	F8.3
2	9 - 16	Latitude (decimal degrees)	F8.3
3-14	17 - 112	Cross-validation errors (absolute values) of Monthly Total Precipitation (mm)	12F8.1

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